

SESSION 1

Basic Physics and Electronics in Radiology

Electrostatics and the X-ray Tube

Electric charge, voltage, electron motion and the physical basis of X-ray production

Course	RAD-201 Radiation Physics
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Lecturer	A. A. Zubair
Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

1. explain electric charge and the behaviour of charged particles;
2. distinguish conductors, insulators and semiconductors using radiography examples;
3. explain electric field, potential difference, current and resistance in simple terms;
4. state Coulomb's law and recognize the inverse-square relationship;
5. describe thermionic emission, space charge and electron acceleration in the X-ray tube;
6. distinguish filament current, tube current, kVp, mA and mAs; and
7. connect basic electrical physics to X-ray quantity, beam energy and safe equipment operation.

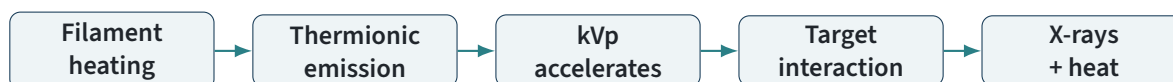
The radiography idea to remember

An X-ray tube is an electrical energy-conversion device. The filament **releases electrons**, the high voltage **accelerates electrons**, and the target interaction converts their kinetic energy mainly into **heat** and partly into **X-rays**. Understanding charge, voltage and current therefore explains what the exposure controls are physically doing inside the tube.

1 Session Orientation: follow the electron

The easiest way to understand this session is to follow one electron through the X-ray system. Four questions organize the whole topic:

1. **Where does the electron come from?** It already exists in the tungsten filament and is released by heating.
2. **What makes it move?** A large potential difference creates an electric field between cathode and anode.
3. **What determines its energy?** The applied tube voltage, expressed clinically as kVp.
4. **What determines how many electrons move?** Filament heating and tube current, expressed as mA and, over time, mAs.



Radiographer's viewpoint

When you select kVp, mA and exposure time at the console, you are controlling this electron pathway indirectly. Good technique selection begins with understanding what each control changes physically, not merely memorizing whether a number should go up or down.

2 Electric charge: the starting point

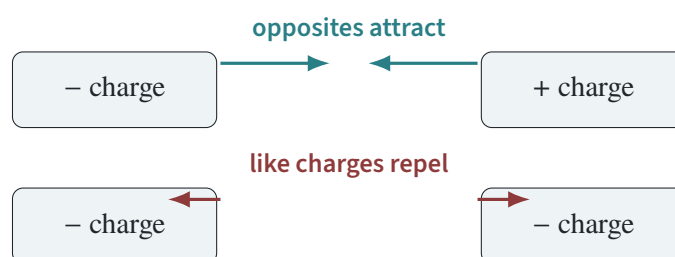
Atoms contain positively charged protons and negatively charged electrons. In ordinary electrical circuits, protons remain bound in atomic nuclei; the particles that move through conductors are electrons.

An object is **electrically neutral** when its positive and negative charges balance. If it gains electrons it becomes negatively charged; if it loses electrons it becomes positively charged. Charge is measured in **coulombs (C)**. The magnitude of the charge on one electron is approximately

$$e = 1.60 \times 10^{-19} \text{ C.}$$

2.1 Attraction, repulsion and conservation of charge

Opposite charges attract; like charges repel. Charge is also conserved: electrical processes move charge from one place to another rather than creating charge from nothing.



Why this matters inside the tube

The negatively charged cathode repels electrons, while the positively charged anode attracts them. The same simple attraction/repulsion rule is the basis of electron acceleration across the evacuated X-ray tube.

3 Conductors, insulators and semiconductors

Different materials control electron motion in different ways.

Material	Electron behaviour	Examples	Radiography relevance
Conductor	Charge moves relatively easily	Copper, aluminium	Cables, connections, transformer windings, tube circuitry
Insulator	Charge movement is strongly restricted	Rubber, glass, ceramic, insulating oil	Prevents leakage and arcing around high-voltage components
Semiconductor	Conductivity can be deliberately controlled	Silicon, selenium	Digital detector electronics and direct/indirect flat-panel systems

Digital radiography connection

Basic electrical materials extend beyond the X-ray tube. Conductors move charge, insulators keep high voltage where it belongs, and semiconductors allow detector systems to convert or process the information carried by X-ray exposure.

4 How objects become charged

Three basic mechanisms are useful for understanding static electricity:

Friction

Contact and rubbing can transfer electrons between materials.

Conduction

Charge is transferred through direct contact with a charged body.

Induction

A nearby electric field redistributes charge without direct contact.

These examples are not the main purpose of radiography physics. The important lesson is that **electrons respond to electrical forces and fields**. That same behaviour is deliberately controlled in generators, X-ray tubes and electronic detectors.

5 Coulomb's law and the inverse-square pattern

For two point charges, the magnitude of the electrostatic force is

$$F = k \frac{|Q_1 Q_2|}{r^2},$$

where Q_1 and Q_2 are the charges, r is their separation and $k \approx 9.0 \times 10^9 \text{ N m}^2/\text{C}^2$.

The radiography lesson is the **inverse-square dependence**. If separation doubles, the force becomes one-quarter:

$$r \rightarrow 2r \quad \Rightarrow \quad F \rightarrow \frac{1}{4}F.$$

Coulomb's law is not the radiation inverse-square law

Both equations contain a $1/r^2$ relationship, but they describe different physical quantities. Coulomb's law describes electrical force between charges. The radiation inverse-square law describes how intensity from an approximately point-like source spreads with distance.

Distance and X-ray intensity

If the beam intensity is 100 units at 1 m, the ideal inverse-square estimate at 2 m is

$$I_2 = 100 \left(\frac{1}{2} \right)^2 = 25 \text{ units.}$$

This is why increasing distance is such an effective radiation-protection measure and why changes in source-to-image distance affect receptor exposure when other factors are unchanged.

6 Electric field and potential difference

An **electric field** is a region in which a charged particle experiences an electrical force. A useful way to picture it is as the “electrical push or pull” available at each location.

Potential difference, or voltage, describes how much electrical energy is transferred per unit charge:

$$V = \frac{W}{Q}.$$

One volt means one joule of energy per coulomb of charge. In diagnostic radiography, the potential across the X-ray tube is thousands of volts, so it is expressed in kilovolts.

From volts to electron energy

An electron accelerated through a potential difference of 1 kV gains approximately 1 keV of kinetic energy. Therefore, an electron accelerated through 100 kV can gain up to about 100 keV before striking the target.

7 Current, resistance and conventional current

Electric current is the rate of flow of charge:

$$I = \frac{Q}{t}.$$

One ampere means one coulomb of charge passing a point each second. Tube current is much smaller and is therefore expressed in milliamperes (mA).

Resistance describes opposition to current flow. Conductors have relatively low resistance, while insulators have very high resistance. In later generator physics, voltage, current and resistance are connected by Ohm’s law.

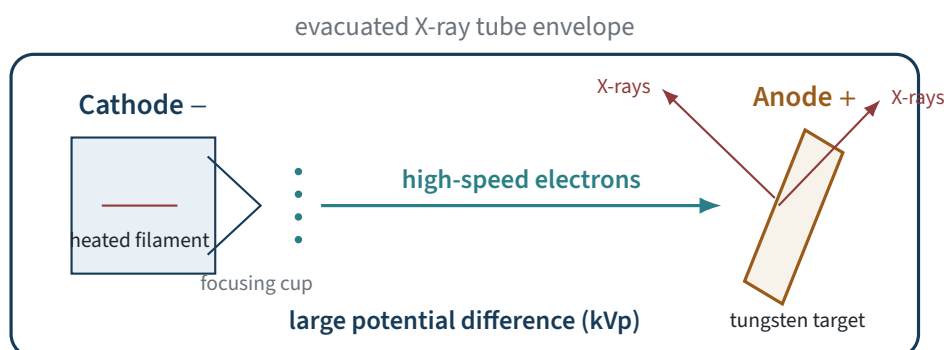
Quantity	Simple meaning	Radiography example
Voltage	electrical energy difference / “push”	kVp across cathode and anode
Current	rate of charge flow	tube current in mA
Resistance	opposition to current	filament and circuit components

Electron flow versus conventional current

Electrons move from the negative side toward the positive side. By historical convention, circuit current is drawn in the opposite direction, from positive to negative. In X-ray-tube discussions, it is usually clearest to state explicitly whether you mean **electron flow** or **conventional current**.

8 The X-ray tube: electrostatics in action

The X-ray tube contains two main electrodes inside an evacuated envelope: a negative **cathode** and a positive **anode**. The vacuum lets electrons travel across the tube with minimal collision with gas molecules.



8.1 Step 1: filament heating and thermionic emission

A separate low-voltage, high-current filament circuit heats the tungsten filament. Heating gives some electrons enough thermal energy to escape the metal surface. This release of electrons is called **thermionic emission**.

The filament does not manufacture electrons. The electrons were already present in the tungsten; heating simply makes a fraction available to leave the surface.

8.2 Step 2: space charge and focusing

Released electrons collect near the filament as a negative **space charge**. Because electrons repel one another, this cloud can limit additional emission unless the tube voltage draws electrons away. The negatively charged focusing cup helps confine the electron cloud and direct it toward the focal spot on the anode.

8.3 Step 3: high-voltage acceleration

When the exposure is made, the generator applies a large potential difference between cathode and anode. The electric field accelerates electrons across the tube. Their maximum kinetic energy is approximately

$$KE_{\max} = eV.$$

Thus, at 80 kVp the highest-energy incident electrons can reach about 80 keV; at 120 kVp they can reach about 120 keV.

8.4 Step 4: target interaction

The electrons strike the tungsten target and decelerate rapidly. In a diagnostic X-ray tube, roughly **99% of their kinetic energy becomes heat and about 1% becomes X-radiation**. The exact efficiency changes with target material and electron energy, but the central practical fact is that heat management is essential.

Why the anode rotates

Many general-radiography tubes use a rotating anode so that heat is spread over a larger focal track. The purpose is not to increase electron attraction; it is primarily to manage the large amount of heat produced during electron-target interactions.

9 kVp, mA, exposure time and mAs

The exposure factors control different parts of the electron story.

Factor	What it controls physically	Main beam effect	Radiography interpretation
kVp	maximum electron energy across the tube	beam energy/penetrability; also influences output	higher kVp generally produces a more penetrating beam
mA	electron flow rate across the tube	photon production rate / quantity	higher mA means more electrons striking target each second
Time	how long current flows	duration of photon production	longer exposure gives more total photons and more motion opportunity
mAs	$\text{mA} \times \text{time}$	total electron quantity during exposure	strong control of total X-ray quantity

Filament current is not tube current

Filament current heats the filament and is usually several amperes at low voltage. **Tube current** is the much smaller electron current crossing from cathode to anode and is displayed in mA. The generator adjusts filament heating to obtain the selected tube current.

mA, time and mAs

Two techniques use the same kVp:

$$200 \text{ mA} \times 0.10 \text{ s} = 20 \text{ mAs}, \quad 400 \text{ mA} \times 0.05 \text{ s} = 20 \text{ mAs}.$$

In the simplified mAs relationship, both deliver the same total tube-current-time product, but the second uses a shorter exposure time and may reduce motion blur. Tube loading and equipment limits still have to be respected.

Digital radiography caution

In digital radiography, displayed image brightness is heavily influenced by processing and is not a reliable visual indicator of patient exposure. Technique evaluation should therefore use exposure indicators, image noise, clinical adequacy and dose-awareness rather than assuming that a normally bright image must have received an appropriate exposure.

10 How kVp and mA differ in the X-ray tube

Students often confuse electron *energy* with electron *number*. Keep the distinction explicit:

- Increasing **kVp** increases the potential difference, so individual electrons can reach the anode with greater kinetic energy. This raises maximum photon energy and usually increases beam penetrability and X-ray output.
- Increasing **mA** mainly increases the number of electrons crossing the tube per second. More incident electrons produce more X-ray photons, but the selected kVp still sets the maximum electron energy.

Increase kVp
greater electron energy

Increase mA
more electrons / second

higher maximum photon energy
and generally more penetration

more target interactions
and greater X-ray quantity

Do not say “higher kVp means more electrons”

Higher kVp mainly increases the energy available to each electron. Higher mA mainly increases the rate at which electrons cross the tube. kVp can also increase X-ray quantity because higher-energy electrons generate radiation more efficiently, but the two controls are not interchangeable.

11 Electrical safety around X-ray equipment

The X-ray tube requires tens to hundreds of kilovolts, so the system must control high voltage safely. Important engineering measures include insulation, protective housing, grounding/earthing, interlocks and appropriate maintenance.

The radiographer is not expected to service high-voltage circuits. The practical responsibilities are to operate equipment within its rated limits, recognize abnormal behaviour, avoid bypassing safety systems and report suspected electrical or tube faults promptly.

Examples of abnormal signs

Unexpected arcing sounds, repeated exposure failure, burning odour, damaged high-voltage cables or unusual generator messages require attention according to departmental procedure. Do not continue using equipment when an electrical fault is suspected merely because an image can still be produced.

12 Putting the physics together

The complete sequence for this session is:

1. Electric charge establishes attraction and repulsion.
2. A voltage across the tube creates an electric field.
3. Filament current heats tungsten and releases electrons by thermionic emission.
4. The space charge provides an electron supply near the cathode.
5. kVp accelerates those electrons toward the anode.
6. Tube current determines how many electrons cross per second; exposure time determines how long they flow.
7. Electron-target interactions produce mostly heat and a small proportion of X-rays.
8. The resulting X-ray beam is then shaped by later processes such as filtration, collimation and patient attenuation.

Key Takeaways

- Electrons are the mobile charge carriers central to X-ray production.
- Voltage is electrical energy per unit charge; current is charge flow per unit time.

- Coulomb's law and radiation intensity both show inverse-square mathematics, but they describe different phenomena.
- Thermionic emission releases electrons from a heated filament; the filament does not create them.
- kVp controls the maximum energy gained by tube electrons and strongly affects beam quality and penetrability.
- mA controls electron flow rate, while mAs combines tube current and exposure time to describe total current-time product.
- The X-ray tube converts electron kinetic energy mainly into heat and partly into X-rays.
- Safe high-voltage operation depends on insulation, grounding and equipment protection systems.

Key Terms

Electric charge

Property of matter that produces electrical attraction or repulsion.

Coulomb

SI unit of electric charge.

Electric field

Region in which a charge experiences electrical force.

Potential difference

Electrical energy transferred per unit charge; measured in volts.

Current

Rate of charge flow; measured in amperes.

Thermionic emission

Release of electrons from a heated metal filament.

Space charge

Electron cloud formed near the cathode filament.

kVp

Peak tube potential; major control of electron energy and X-ray beam quality.

mA

Tube current; rate of electron flow across the X-ray tube.

mAs

Product of tube current and exposure time; major determinant of total X-ray quantity.

Self-Assessment Questions

1. Why are electrons, rather than protons, the important mobile charges in ordinary imaging equipment?
2. Distinguish a conductor, insulator and semiconductor and give one radiography example of each.

3. State Coulomb's law and explain what happens to force when separation doubles.
4. Distinguish electric field, potential difference, current and resistance.
5. Explain why an electron accelerated through 100 kV can gain up to about 100 keV of kinetic energy.
6. Describe thermionic emission and the formation of space charge.
7. Distinguish filament current from tube current.
8. Compare the physical effects of increasing kVp and increasing mA.
9. Calculate the mAs for 320 mA at 0.025 s.
10. Why does a diagnostic X-ray tube need effective heat management?

Bridge forward

This session explains how electrons are supplied and accelerated. The next stage of radiation physics is to examine how electrical circuits and the X-ray generator create, control and rectify the voltages and currents required for a safe, reproducible exposure.

